

Volumetric Optical Processing via Sub-Diffraction Implosion Carving: A Monolithic 3D Photonic Architecture for High-Dimensional Inference

Technical Architecture & Feasibility Analysis

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Abstract

We present a volumetric optical processing architecture that maps high-dimensional linear and nonlinear operations onto a monolithic three-dimensional photonic volume fabricated by sub-diffraction Implosion Carving (ImpCarv). ImpCarv enables nanoscale refractive-index programming throughout a hydrogel-derived solid by two-photon vacancy patterning followed by isotropic shrinkage greater than tenfold linearly. Building on this capability, we define a bound single-mode waveguide network that routes phase- and amplitude-encoded optical payloads to cured metallic and doped terminal nodes. Linear transformations are realized by interference at these nodes; passive nonlinear dynamics—reverse saturable absorption for intensity thresholding, excited-state lifetime decay for short-term temporal kernels, and cross-phase modulation for field-field interaction—supply activation, sequence context, and dynamic operand products. Computation occurs during optical time-of-flight through the crystal. We detail the end-to-end compilation and fabrication workflow, provide a first-order materials and energy cost model with parametric yield sensitivity, and identify the multiphysics parameters that govern cascadability and fidelity. The architecture offers a path toward dense, largely passive optical inference engines that exploit the full three-dimensional volume rather than planar or free-space diffractive layers alone.

1. Introduction

Optical computing has long promised high bandwidth, massive spatial parallelism, and low energy per operation by performing linear algebra in the analog optical domain. Diffractive deep neural networks and integrated photonic tensor cores have demonstrated classification and matrix multiplication at the speed of light, yet both families of systems face structural limits. Free-space diffractive volumes suffer from diffraction and modal crosstalk when scaled; planar silicon-photonic circuits remain largely two-dimensional and rely on active phase shifters or heaters whose power and density constrain large models.

A complementary route is to program the refractive-index landscape throughout a solid three-dimensional volume at sub-wavelength resolution. Recent progress in Implosion Carving (ImpCarv) provides exactly this capability: two-photon photopatterning of a swollen hydrogel followed by isotropic dehydration yields feature sizes below 100 nm and refractive-index contrast sufficient for visible-wavelength phase control. A simple all-optical classifier has already been realized with this process.

Here we develop the architectural consequences of that fabrication primitive. We define a monolithic three-dimensional photonic circuit in which an internal single-mode waveguide network routes optical fields to localized terminal nodes that execute linear transformations, while co-located nonlinear and lifetime-engineered regions supply the activation, temporal, and multiplicative dynamics required for more expressive models. The resulting system is intended to perform inference during a single optical transit through a centimeter-scale crystal, collapsing the dominant latency to time-of-flight.

2. Fabrication Basis: Implosion Carving

ImpCarv begins with a water-rich polyacrylate hydrogel precursor. A scanned two-photon laser generates reactive oxygen species that cleave the scaffold at programmed locations, creating vacancies with an initial written resolution of approximately 800 nm. Subsequent ionic dehydration collapses the gel isotropically by more than tenfold linearly (approximately 2,000-fold by volume). Supercritical CO₂ drying preserves the resulting sub-100 nm geometries. The refractive-index contrast between air-filled vacancies and the dehydrated scaffold reaches ~0.5, enabling precise phase control at visible wavelengths.

The same process can incorporate functional materials. Two-photon chemical reduction or cross-linking selectively cures metallic precursors (TiO₂, silver, gold) or high-index chromophores at designated coordinates, forming terminal scattering and reflection nodes after shrinkage. Photosensitizers, reverse-saturable-absorber dopants, and lifetime-engineered chromophores may be introduced into the precursor matrix and spatially localized by the same writing step. These extensions convert a passive refractive-index metastructure into a hybrid linear–nonlinear volumetric circuit.

3. Architecture

3.1 Monolithic 3D Optical Bus

Phase- and amplitude-encoded optical payloads are launched from an external spatial light modulator or interferometric source into face-coupled input ports. Inside the crystal, a continuous network of single-mode waveguides—patterned as micro-channels prior to shrinkage—guides the fields along non-intersecting three-dimensional paths to terminal nodes. Bound propagation eliminates free-space beam divergence and K-space degeneracy, preserving mode structure and relative phase for subsequent

interference and detection.

3.2 Terminal Nodes and Linear Transformations

Weight matrices are realized when guided waves encounter two-photon-cured terminal nodes. These nodes function as static beam splitters, phase shifters, and reflective diffractive elements whose complex scattering amplitudes are set by geometry and material composition. Because the inputs arrive via single-mode guides rather than unguided fields, the resulting interference is deterministic and free from the spatial degradation characteristic of bulk diffractive volumes. Homodyne detection at the output face recovers the transformed optical field.

3.3 Passive Nonlinear and Temporal Dynamics

While the linear weight landscape is fixed by fabrication, runtime adaptivity is obtained from intrinsic material nonlinearities without electronic control loops.

Intensity thresholding via reverse saturable absorption. Zones co-doped with reverse-saturable absorbers remain strongly absorbing at low intensity. Above a threshold set by the ratio of excited-state to ground-state absorption cross-sections, the medium bleaches and transmits. This intensity-dependent transmission supplies a passive, event-driven activation or sparsity function whose threshold can be tuned spatially by local dopant concentration.

Short-term temporal kernel via excited-state lifetime. Optical excitation of selected chromophores induces temporary shifts in refractive index and absorption that relax exponentially with material lifetime τ . Sequential pulses arriving within a window of order τ therefore encounter a modified medium, implementing a deterministic exponential decay kernel. Such kernels are mathematically equivalent to the continuous-time memory operators used in state-space and linear recurrent models, providing sequence context without external memory.

Field–field interaction via cross-phase modulation. In regions of elevated third-order susceptibility—achieved by high- $\chi^{(3)}$ chromophores or localized nanoparticle enhancement—overlapping optical fields modulate one another’s phase. This cross-phase interaction supplies a physical substrate for dynamic operand products, including elements of attention, without requiring reconfigurable Mach–Zehnder arrays.

4. End-to-End Workflow

The compilation and fabrication sequence proceeds in four phases.

Phase 1 — Topology synthesis. Model weights are extracted and mapped onto a three-dimensional waveguide interconnect graph and terminal-node coordinate set. An inverse shrinkage deformation is applied so that the written geometry, after isotropic collapse, matches the design.

Phase 2 — Material preparation and writing. The polyacrylate precursor is doped with photosensitizers, metal or high-index precursors, reverse-saturable-absorber chromophores, and lifetime-engineered dyes. Two-photon laser scribing defines both the waveguide channels and the terminal-node locations.

Phase 3 — Implosion and drying. Ionic dehydration collapses the scaffold; supercritical CO₂ drying locks the sub-100 nm features in place.

Phase 4 — Assembly and execution. The monolithic crystal is mounted in an alignment housing. Phase/amplitude payloads are launched into the input ports; the transformed fields are read out at a sensor array by homodyne or direct detection.

5. Materials and Energy Cost Model

A first-order bill-of-materials estimate based on laboratory-scale precursor volumes, modest laser energy, and idealized process conditions yields a materials-plus-energy floor of approximately \$24 per crystal (Table 1). The dominant contributors are the custom RSA chromophores and the hydrogel base; laser energy and dehydration energy are negligible at current laboratory rates. This figure is a nominal lower bound. It does not include capital amortization of the writing system, volumetric write time at scale, packaging, alignment, testing, or yield. Sensitivity analysis against defect density, write speed, and alignment tolerance is required before economic projections can be refined.

Component / Process	Function	Quantity	Est. Cost (USD)
Polyacrylate hydrogel base	Swellable 3D scaffold	30 mL	\$1.20
TiO ₂ / metal precursors	Terminal node reflectors	<0.5 g	\$0.35
RSA chromophores	Nonlinear gating	Micro-dosing	\$18.50
Photoinitiators & solvents	Patterning & shrinkage bath	1 L bath	\$3.50

Femtosecond laser energy	Waveguide & node writing	0.4 kWh	\$0.08
Dehydration / drying energy	Thermal + supercritical	1.5 kWh	\$0.30
Nominal materials + energy floor			~\$23.93 / crystal

Table 1. First-order materials and process-energy cost model. Values reflect laboratory-scale precursor pricing and idealized yield; capital, packaging, and yield adjustments are excluded.

5.1 Parametric Sensitivity and Yield Modeling

The nominal bill-of-materials floor of \$23.93 per core assumes ideal yields, zero tool overhead, and instantaneous lithographic writing. To evaluate realistic commercial economics, unit cost C_{unit} is modeled as a function of raw materials, tool operational amortization, volumetric patterning throughput, and compound processing yields:

$$C_{\text{unit}} = (C_{\text{BOM}} + t_{\text{write}} \cdot R_{\text{tool}} + C_{\text{pkg}}) / (Y_{\text{fab}} \cdot Y_{\text{pkg}})$$

where $C_{\text{BOM}} = \$23.93$ is the baseline material and energy floor; t_{write} is two-photon patterning time per crystal (hours), derived from active patterned volume V_{pattern} (mm^3) and volumetric write speed v_{write} (mm^3/s); R_{tool} is the fully burdened tool operational rate (\$/hr); $Y_{\text{fab}} = \exp(-D_0 \cdot V_{\text{core}})$ is volumetric yield governed by spatial defect density D_0 (defects/ cm^3); and Y_{pkg} is mechanical alignment and fiber-coupling packaging yield.

Parameter / Metric	Pilot Scale	Production Target	Scaled Ideal
Volumetric write speed v_{write}	0.10 mm^3/s	5.00 mm^3/s	10.00 mm^3/s
Write time per core t_{write}	10.00 hrs	0.20 hrs	0.10 hrs
Tool operational rate R_{tool}	\$150 / hr	\$75 / hr	\$50 / hr
Volumetric defect density D_0	0.50 / cm^3	0.05 / cm^3	0.01 / cm^3
Fabrication yield Y_{fab}	60.7%	95.1%	99.0%
Packaging & alignment yield Y_{pkg}	75.0%	92.0%	98.0%
Packaging & housing cost C_{pkg}	\$120	\$35	\$15
Compound yield Y_{total}	45.5%	87.5%	97.0%
Effective unit cost C_{unit}	\$3,611	\$84.50	\$45.28

Table 2. Parametric cost model across pilot, production-target, and scaled-ideal regimes. Write-speed and defect-density improvements dominate the transition from laboratory to commercial unit cost.

Primary cost drivers. Single-beam direct laser writing at current pilot speeds (0.10 mm^3/s) leaves tool amortization as the dominant cost term. Reaching unit economics below \$100 requires multibeam holographic or SLM parallelization to achieve $v_{\text{write}} \geq 5 \text{ mm}^3/\text{s}$. In addition, unlike planar ICs where local defects disable individual devices, a sub-micron inclusion inside a 3D single-mode channel can disrupt downstream phase coherence across the interference network. High-purity precursor filtration and supercritical drying cleanliness are therefore mandatory to keep $D_0 \leq 0.05 \text{ defects}/\text{cm}^3$.

6. Discussion and Outlook

The architecture rests on three linked capabilities: (i) nanoscale three-dimensional refractive-index programming via ImpCary, (ii) bound single-mode routing that replaces free-space propagation, and (iii) localized passive nonlinearities that furnish activation, temporal context, and field-field products inside the same volume. Together they address density, functional completeness, and the desire to keep inference largely passive.

Cascadability remains the central systems constraint. Each waveguide segment, terminal node, RSA zone, and cross-phase-modulation region contributes insertion loss and phase error. Cumulative optical signal-to-noise ratio and phase budget therefore set the maximum depth of computation that can be performed in a single transit without intermediate amplification. Quantifying waveguide loss after shrinkage, node scattering efficiency, RSA extinction ratio and recovery, and the achievable cross-phase shift per unit power constitutes the immediate multiphysics agenda.

If those bounds prove favorable, the platform offers a route to optical inference engines whose compute density and energy profile differ markedly from both planar photonic integrated circuits and free-space diffractive stacks. The same fabrication sequence that writes the linear weight landscape can also place the nonlinear and temporal elements, suggesting a path toward truly monolithic volumetric processors.

7. References

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Appendix A: FDTD Multiphysics & Material Simulation Parameters

Full-wave propagation through the monolithic volume is modeled by coupling Maxwell's equations on a Yee-cell grid to two-level rate equations and Kerr polarization terms. Table A1 summarizes the nominal baseline and target operational ranges for the principal material and waveguide parameters.

Mechanism / Domain	Parameter	Symbol	Baseline	Target Range
Optical carrier	Vacuum wavelength	λ_0	532 nm	532–808 nm
Linear substrate	Scaffold base index	n_0	1.48	1.45–1.52
Waveguide confinement	Peak index contrast	Δn	0.40	0.20–0.50
Propagation loss	Waveguide attenuation	α	0.25 dB/cm	0.10–0.50 dB/cm
Terminal node routing	Node insertion loss	L_{node}	0.08 dB/node	0.05–0.20 dB/node
Cross-phase modulation	3rd-order susceptibility	$\chi^{(3)}$	$1.5 \times 10^{-17} \text{ m}^2/\text{V}^2$	10^{-18} – 10^{-16}
Kerr nonlinearity	Nonlinear index	n_2	$3.2 \times 10^{-11} \text{ cm}^2/\text{W}$	10^{-12} – 10^{-10}
RSA activation	Ground-state cross-section	σ_0	$2.0 \times 10^{-18} \text{ cm}^2$	10^{-18} – 10^{-17}
RSA activation	Excited-state cross-section	σ_{ex}	$8.0 \times 10^{-17} \text{ cm}^2$	5×10^{-17} – 5×10^{-16}
RSA dynamic range	Cross-section ratio	$\sigma_{\text{ex}}/\sigma_0$	40	20–100
Temporal memory	Relaxation lifetime	τ	1.2 ns	100 ps–10 ns

Table A1. Nominal baseline and target ranges for FDTD multiphysics parameters.

Solver directives. Spatial discretization must satisfy $\Delta x, \Delta y, \Delta z \leq \lambda_0 / (20 \cdot n_{\text{max}}) \approx 14 \text{ nm}$ to resolve sub-100 nm post-shrinkage features without numerical dispersion. Maximum computational depth per unamplified crystal is constrained by cumulative insertion loss $\Sigma(L_{\text{node}} + \alpha \cdot d) \leq 18 \text{ dB}$, maintaining OSNR $\geq 22 \text{ dB}$ at the homodyne detection array.

Appendix B: Optical Derivation of XPM-Driven Dynamic QK^T Attention

Query and key vectors $\mathbf{Q}, \mathbf{K} \in \mathbb{C}^d$ are encoded as complex optical fields propagating in single-mode volumetric channels. For modes i and j the fields entering a localized $\chi^{(3)}$ interaction node are

$$E_{Q,i} = A_0 Q_i \exp(i(\omega_0 t - k_0 z + \phi_{Q,i}))$$

$$E_{K,j} = A_0 K_j \exp(i(\omega_0 t - k_0 z + \phi_{K,j}))$$

with $A_0 = \sqrt{(2P_0 / (n_0 c \epsilon_0))}$ and $k_0 = 2\pi n_0 / \lambda_0$.

At the interaction node the fields co-propagate over effective length L_{eff} . The total intensity contains the interference term $2 A_0^2 Q_i K_j \cos(\Delta\phi_{ij})$. Under phase alignment the local refractive-index perturbation induced by the Kerr effect is proportional to intensity, and a co-propagating value field $E_{V,j}$ accumulates a nonlinear phase shift

$$\Delta\Theta_{ij} = \gamma L_{\text{eff}} \cdot 2 P_0 Q_i K_j$$

where $\gamma = 2\pi n_2 / \lambda_0$. A dual-rail balanced configuration isolates the bilinear product, yielding a phase differential $\delta\Theta_{ij} = \kappa Q_i K_j$ with coupling constant $\kappa = 4 \gamma L_{\text{eff}} P_0$.

The phase-modulated value field then passes through a downstream RSA intensity-thresholding zone. For small phase shifts a first-order expansion recovers a term proportional to $Q_i K_j V_j$. Scaling dimension $\sqrt{d_k}$ is set physically by choice of interaction length. Homodyne detection at the output face reconstructs the modulated inner-product contribution. In this construction, elements of dynamic QK^T attention are executed during passive optical transit without electronic memory access or active phase-shifter dissipation.